The labour input channel

A supply side determinant of potential output

PROFESSOR KIRSTEN CORNELSON · GLOBAL MACROECONOMICS FOR MANAGERS

IVEY

In today's Globe and Mail

Congo rolls out Canadian-developed vaccine to fight Ebola outbreak

GEOFFREY YORK
AFRICA BUREAU CHIEF
JOHANNESBURG

Congolese authorities have begun rolling out a Canadian-developed vaccine in an attempt to protect 20,000 health workers from a fast-spreading Ebola outbreak, but Ottawa could be doing more to develop vaccines that are effective against the current species of the virus, experts say.

The vaccine, known as Ervebo and manufactured by Merck, was developed at the National Microbiology Laboratory in Winnipeg and has been deployed in outbreaks of the most common Ebola species, known as the Zaire virus. Its protective value against the Bundibugyo Ebola virus – the cause of the latest outbreak – is unknown, but health authorities have decided to try it anyway in a large-scale study.

Ervebo was developed and licensed solely for the Zaire virus, but it is the only Ebola vaccine available in large quantities today. There are no proven vaccines for Bundibugyo.

A leading Canadian vaccine researcher suggested in 2014 that the basic platform of what became Ervebo could be developed to target the Bundibugyo Ebola virus, documents show. The work was never done, and the international focus today has shifted to potential vaccines from other developers.

The current Ebola outbreak has infected more than 7,670 people and killed nearly 3,700 in the Democratic Republic of the Congo, according to official data Monday. It is already the worst Ebola epidemic in Congo's history and is spreading faster than the world's biggest outbreak, which killed more than 11,300 people in Liberia, Guinea and Sierra Leone from 2014 to 2016.

The outbreak has been fuelled by armed conflicts in eastern Congo that have destabilized the region and forced thousands of people to flee their homes. The lack of a proven vaccine has hampered the response to the crisis.

The vaccination campaign, launched Saturday, will target 20,000 front-line health workers in the epicentre of the outbreak, Congo's Ituri and North Kivu provinces, over the next nine to 12 months to see whether Ervebo is effective against the Bundibugyo Ebola virus.

"Preliminary data suggest that it may offer some degree of protection against the virus," said Guyguy Manangama, an epidemiologist and deputy director of Congo Ebola operations with Médecins Sans Frontières (MSF).

"In a race against time ... we must use every tool available to us," he said in a statement. "Even partial protection could help reduce severe illness and death. However, these potential benefits must be demonstrated scientifically."

Judie Alimonti, a scientist at the Winnipeg lab who helped develop Ervebo, wrote in a 2014 internal report that she strongly recommended the use of the same vaccine platform to create vaccines for the Bundibugyo and other Ebola viruses. But she noted that there was a lack of money to support the research work in Canada.

"The scientists would have happily continued working on it," said Matthew Herder, a professor and associate director of the Health Justice Institute at Dalhousie University in Halifax.

"It's just that they had no more funding from the government, so they couldn't continue to work," he told The Globe and Mail in an interview. "Canada could do more now – and they could have done more in the past."

The 2014 report by Dr. Alimonti was obtained under the Access to Information Act by three researchers, including Prof. Herder.

He said he is concerned that Ottawa's current emphasis on defence spending could reduce its support for health research, including scientific research on Ebola. The government should recognize that diseases such as Ebola are global security issues that could affect Canada, he said.

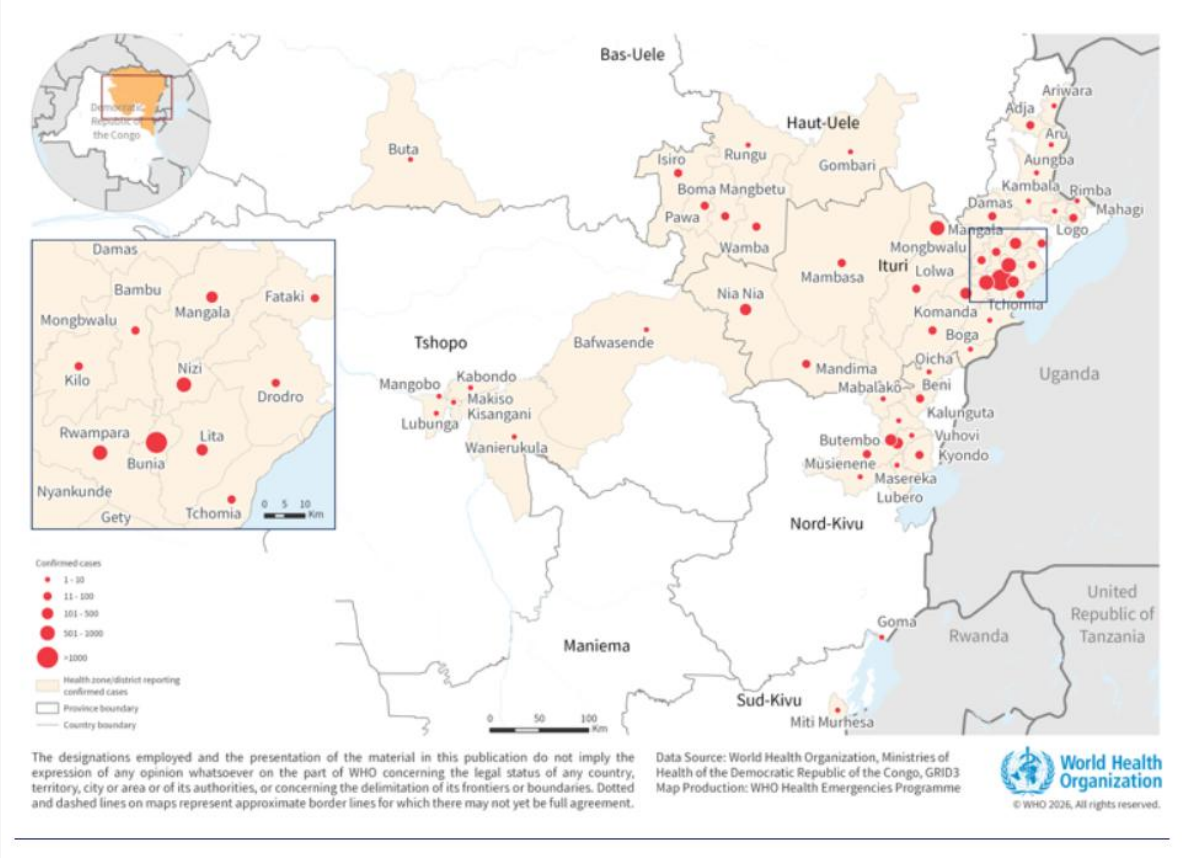
Adam Houston, a medical policy and advocacy adviser at MSF Canada, said the Ervebo trial is a reminder of Canada's potential role in future vaccine development.

"It is really surprising that, despite the effectiveness of the original Ebola vaccine developed in the Government of Canada's own laboratory, the Canadian government has been largely absent from efforts to further develop proven Canadian technology to tackle other viruses causing Ebola disease, including Bundibugyo," he told The Globe.

The Coalition for Epidemic Preparedness Innovations is funding studies of four potential vaccines for Bundibugyo. One of them, developed by Moderna, is undergoing its first clinical trial at three sites in Canada.

Karine LeBlanc, a spokesperson for Health Canada, said the national lab in Winnipeg is continuing to conduct laboratory research on Ebola to develop vaccines, treatments and diagnostic tests, including possible Bundibugyo vaccines.

Preliminary data from an animal study at the lab, presented to the World Health Organization in July, demonstrated that Ervebo gave ferrets full protection from Bundibugyo, she noted.



Three big questions about Proposition 1

PROP 1 In most economies, real output (GDP) follows a rising trend determined by the supply side of the economy.

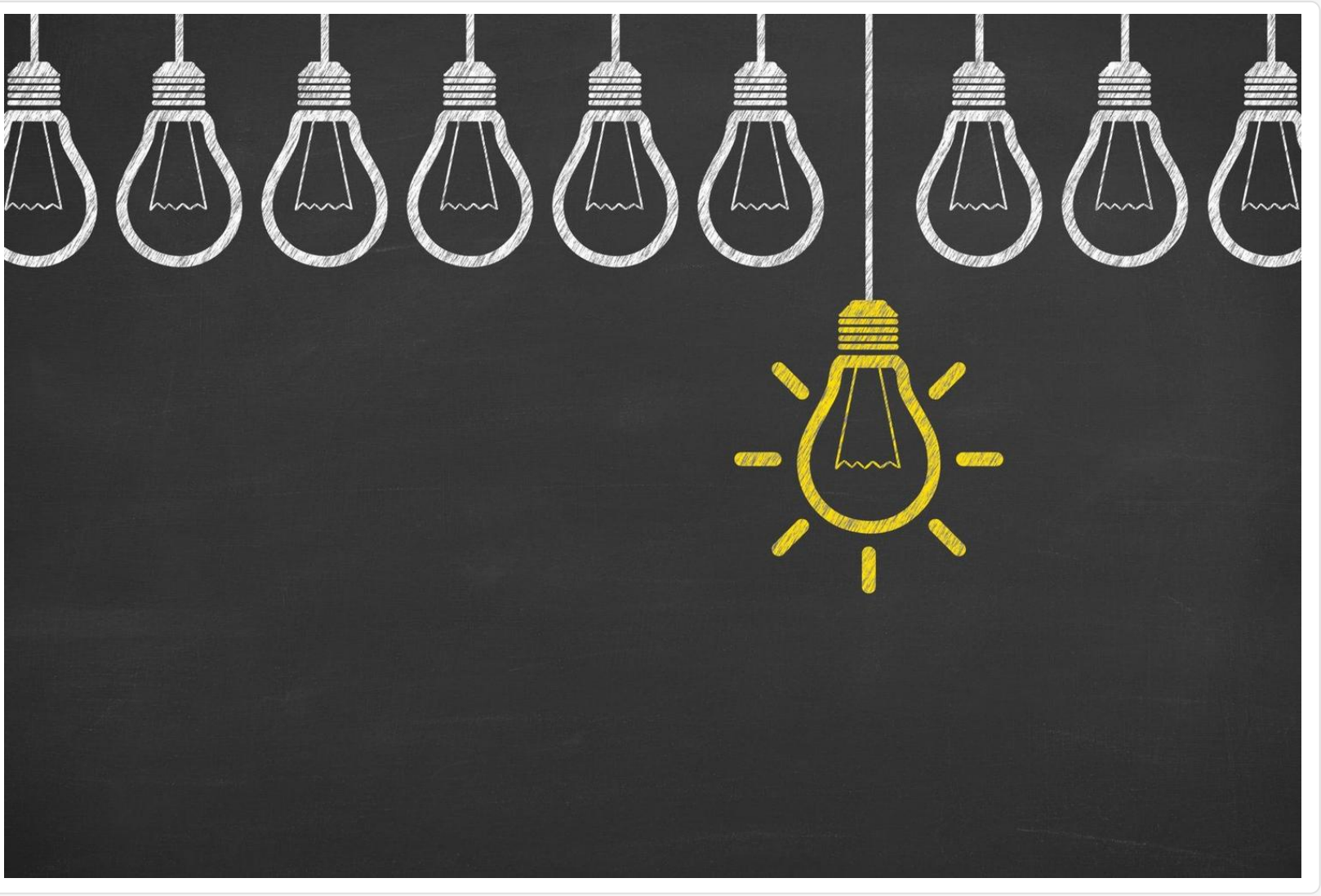
1 Is it true? Do most economies experience rising GDP over time?

2 What's so good about GDP growth anyway?

3 What do we mean when we say trend growth is "supply side determined"?

— Q3 is the core conceptual question for the first half of GMM

Our growth model



Motivation: A simple GDP decomposition

$$GDP = Y$$

Motivation: A simple GDP decomposition

$$Y = \left(\frac{Y}{Population}\right) * Population$$

Motivation: A simple GDP decomposition

$$Y = \left(\frac{Y}{Labour\ Hours}\right) * \left(\frac{Labour\ Hours}{Population}\right) * Population$$

Motivation: A simple GDP decomposition

$$Y = \left(\frac{Y}{Labour\ Hours}\right) * \left(\frac{Labour\ Hours}{Workers}\right)\left(\frac{Workers}{Population}\right) * Population$$

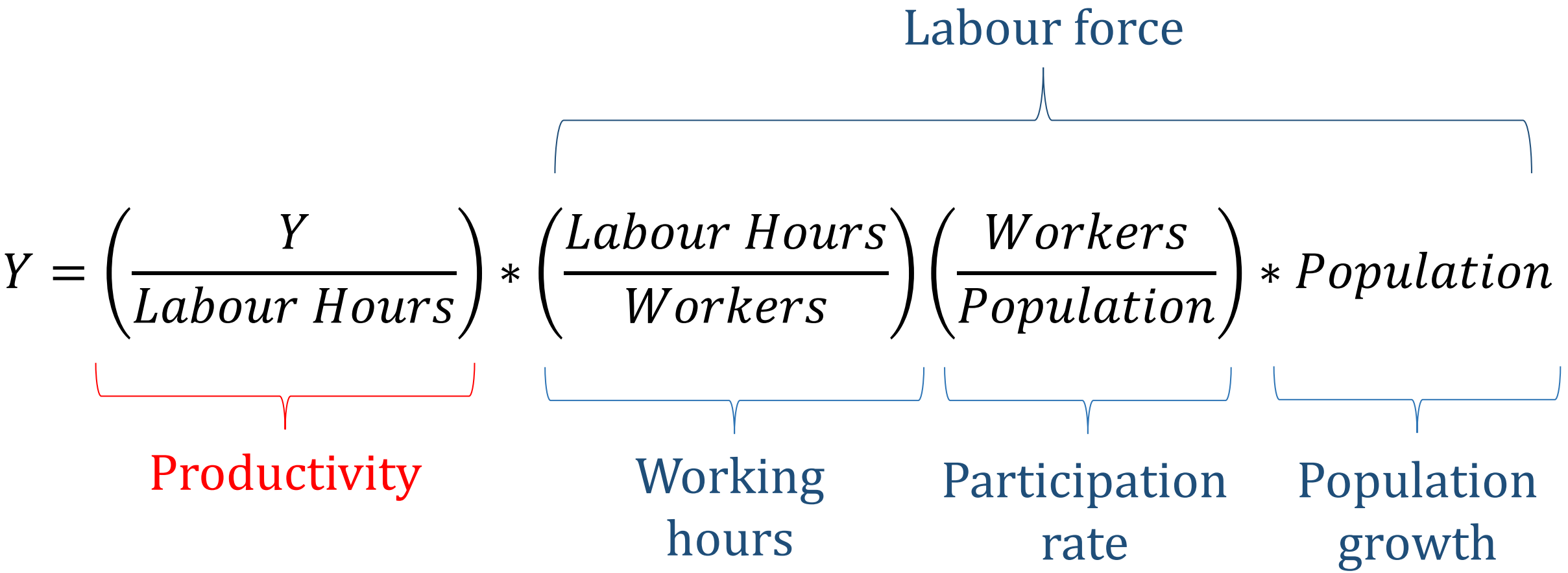
Motivation: A simple GDP decomposition

Labour force

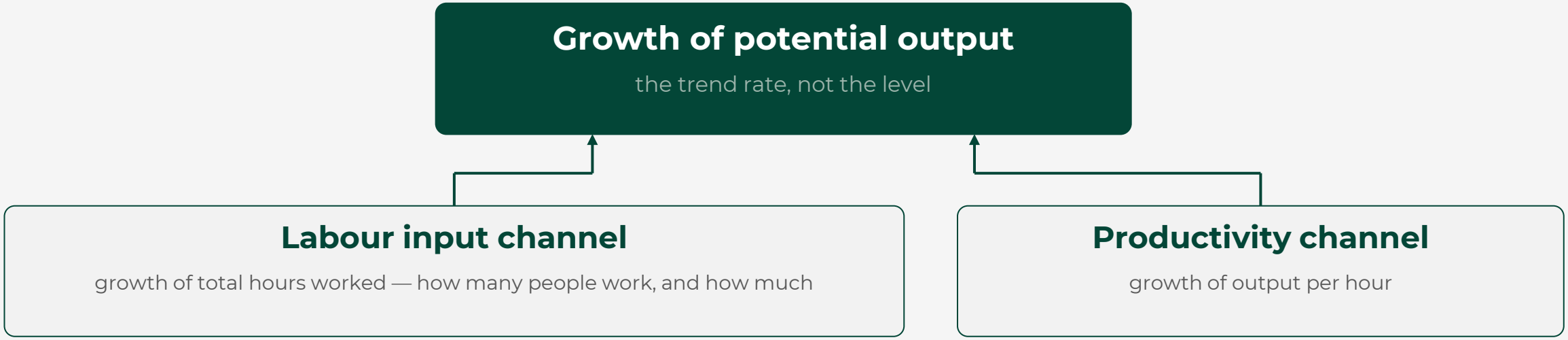
$$Y = \left(\frac{Y}{Labour\ Hours}\right) * \left(\frac{Labour\ Hours}{Workers}\right)\left(\frac{Workers}{Population}\right) * Population$$

Productivity

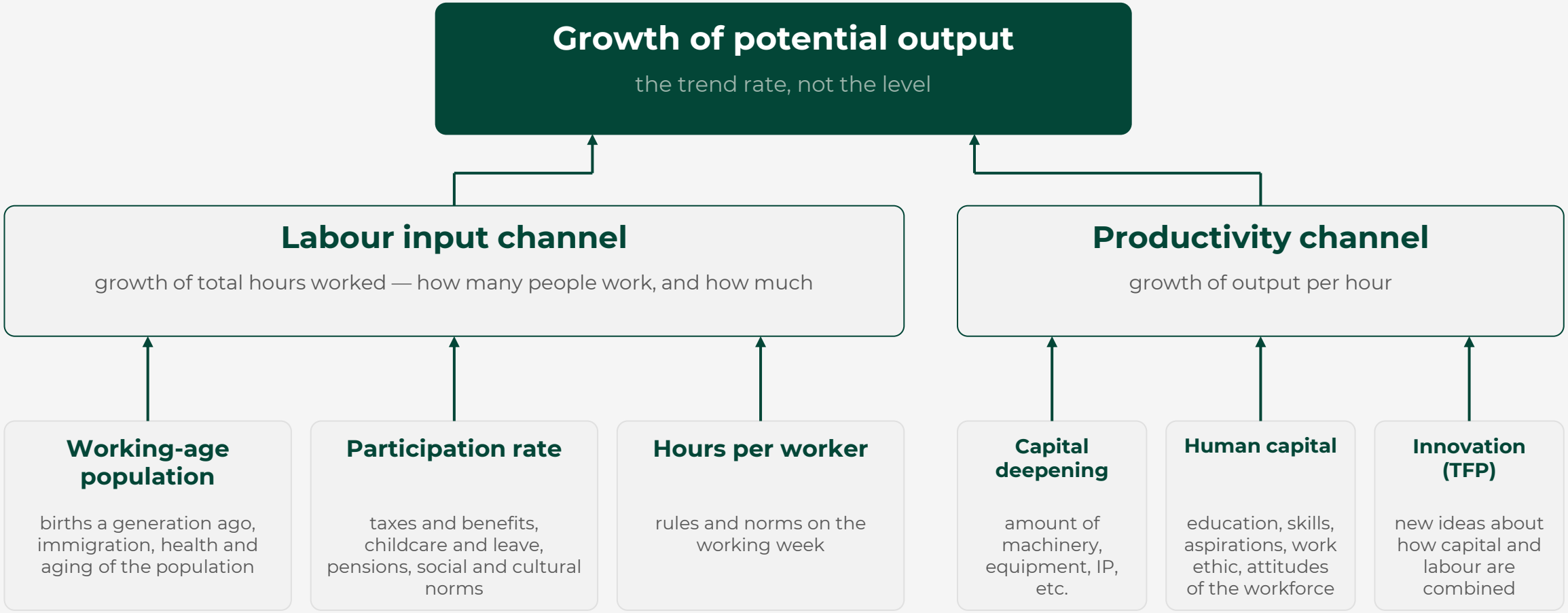
Motivation: A simple GDP decomposition



Our growth model: two fundamental channels



The supply side determinants of potential output



Today’s agenda

1

Trend growth: the labour input channel

Working-age population × participation × hours — the arithmetic of labour input

2

Case study 1: Europe’s hours worked “problem”

Germans work 450 fewer hours than Americans — loss, bargain, or purchase?

3

Case study 2: Who works more, rich people or poor people?

Inside the US: the best paid work the longest — greedy jobs, and who supplies them

Warm-up: Do you understand our growth model?

WORK IN PAIRS

Nine headlines, one at a time.

Real events, lightly rewritten.

Decide, with your partner, where it lands in our growth model.

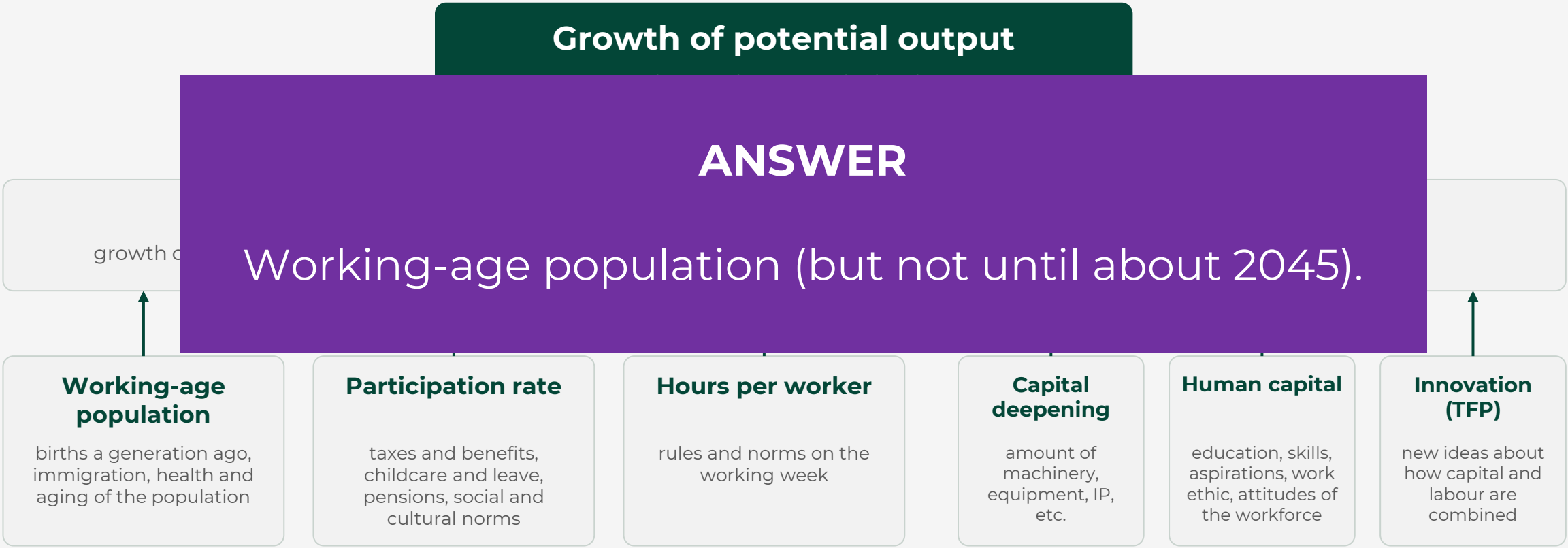
Which box of the labour input channel does the headline move — working-age population, participation rate, or hours per worker? Or does it belong somewhere else in the model — or nowhere in it at all?

Be ready to say why in one sentence.

The box is the easy part. The reason is what we are after.

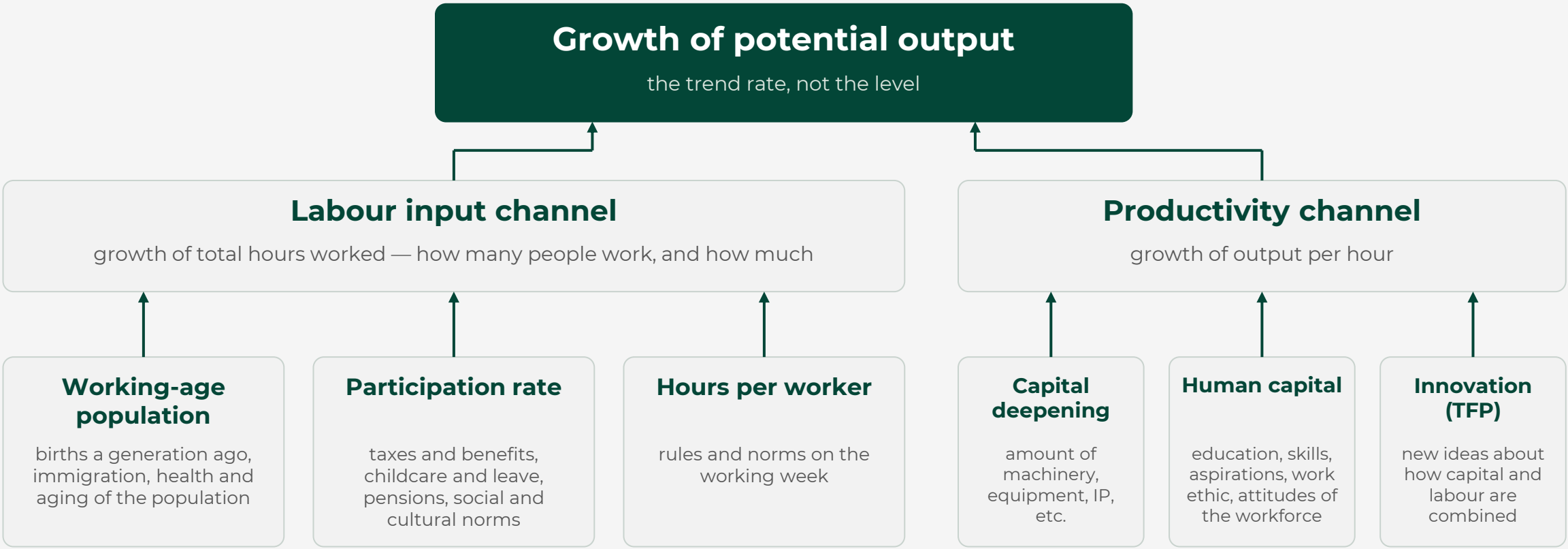
EXAMPLE

“Canada's fertility rate hits record low of 1.25.”



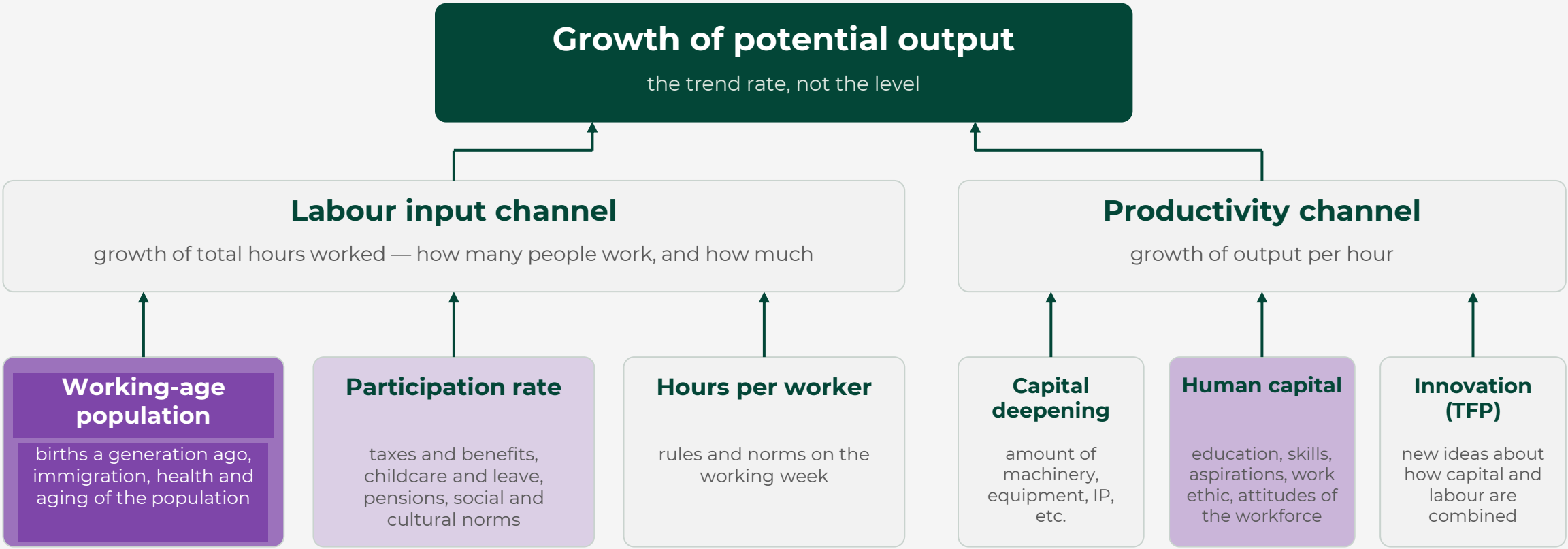
Which box does this move — or is it outside the labour input channel altogether?

“Ottawa halves new study permits for foreign students; colleges cut programs and staff.”



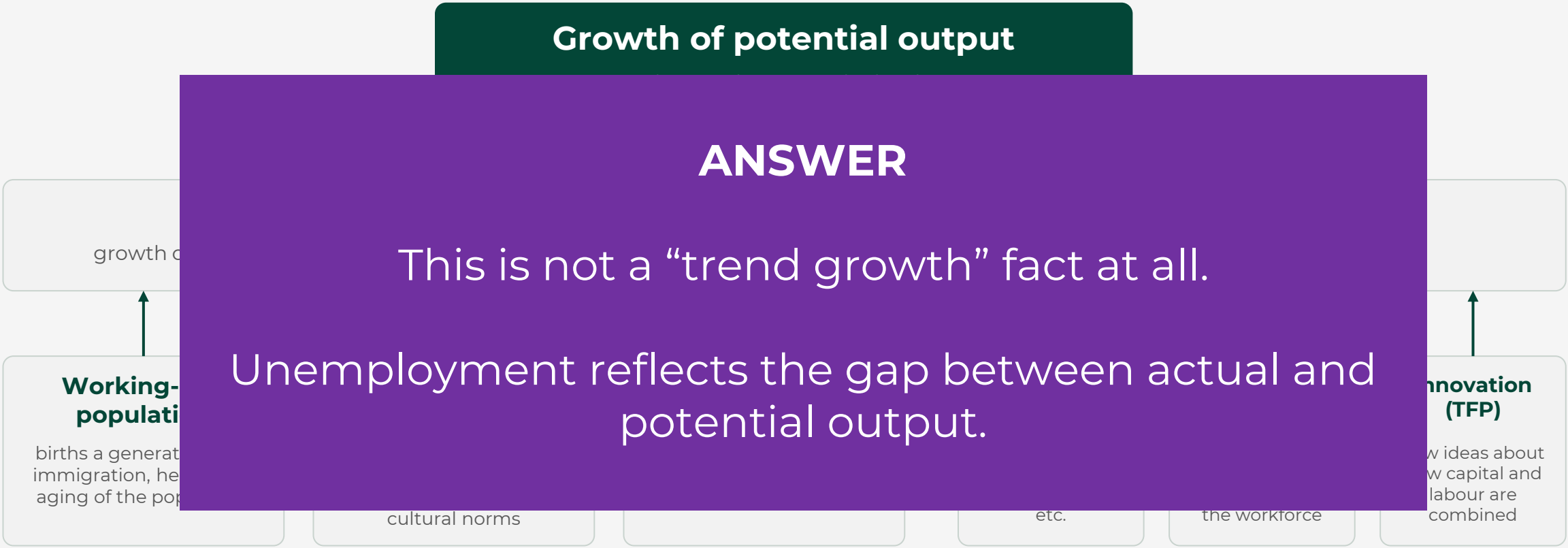
Which box does this move — or is it outside the labour input channel altogether?

“Ottawa halves new study permits for foreign students; colleges cut programs and staff.”



Which box does this move — or is it outside the labour input channel altogether?

“Youth unemployment climbs past 14%, worst summer job market since 2010 outside the pandemic.”



ANSWER

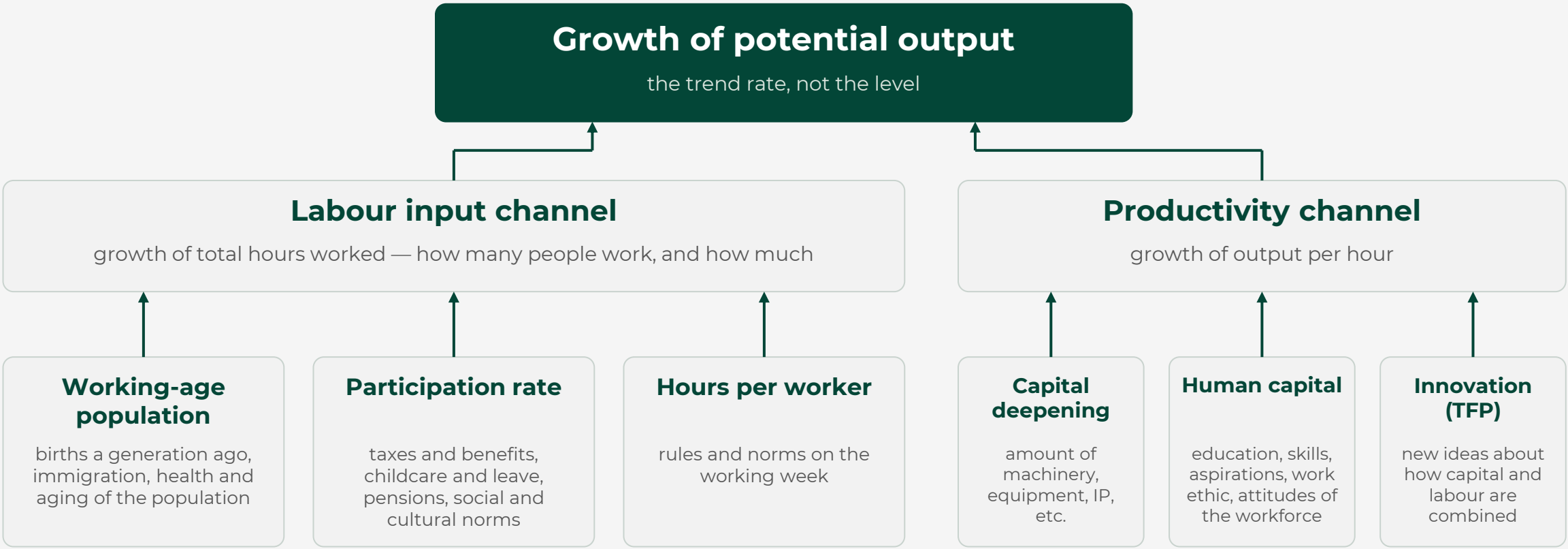
This is not a “trend growth” fact at all.

Unemployment reflects the gap between actual and potential output.

Which box does this move — or is it outside the labour input channel altogether?

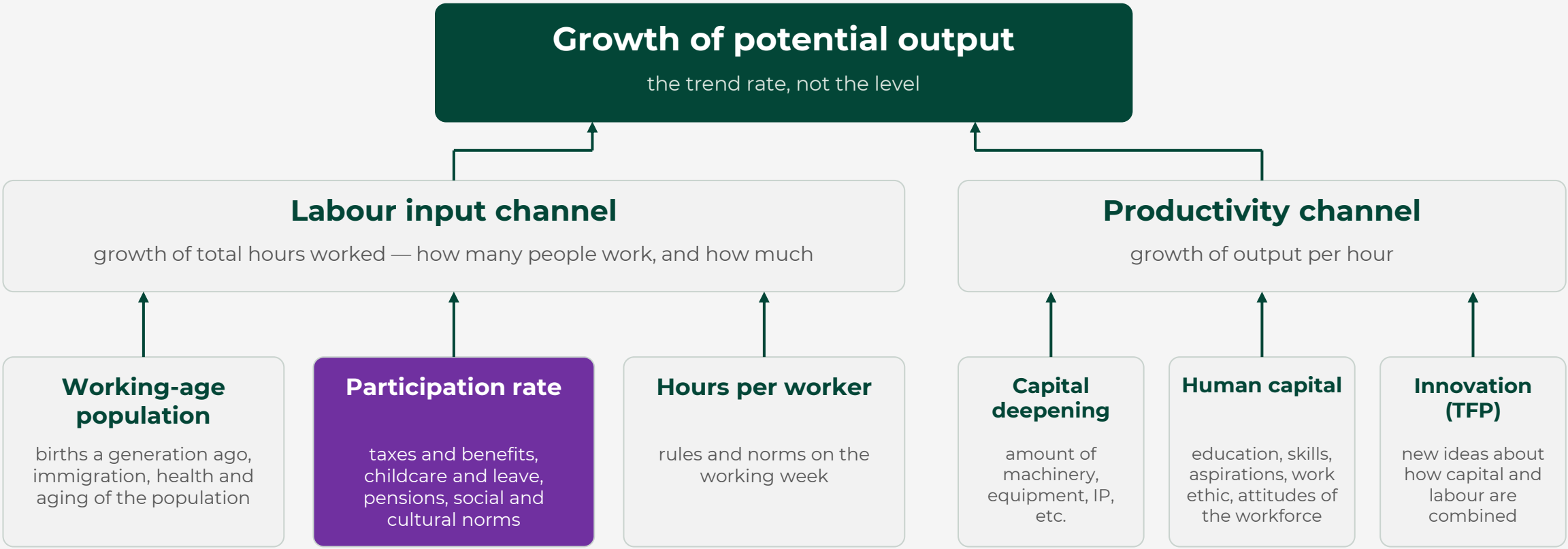
WARM-UP · HEADLINE 4 OF 9

“Working past 65 now adds 8.4 percent a year to your CPP cheque.”



Which box does this move — or is it outside the labour input channel altogether?

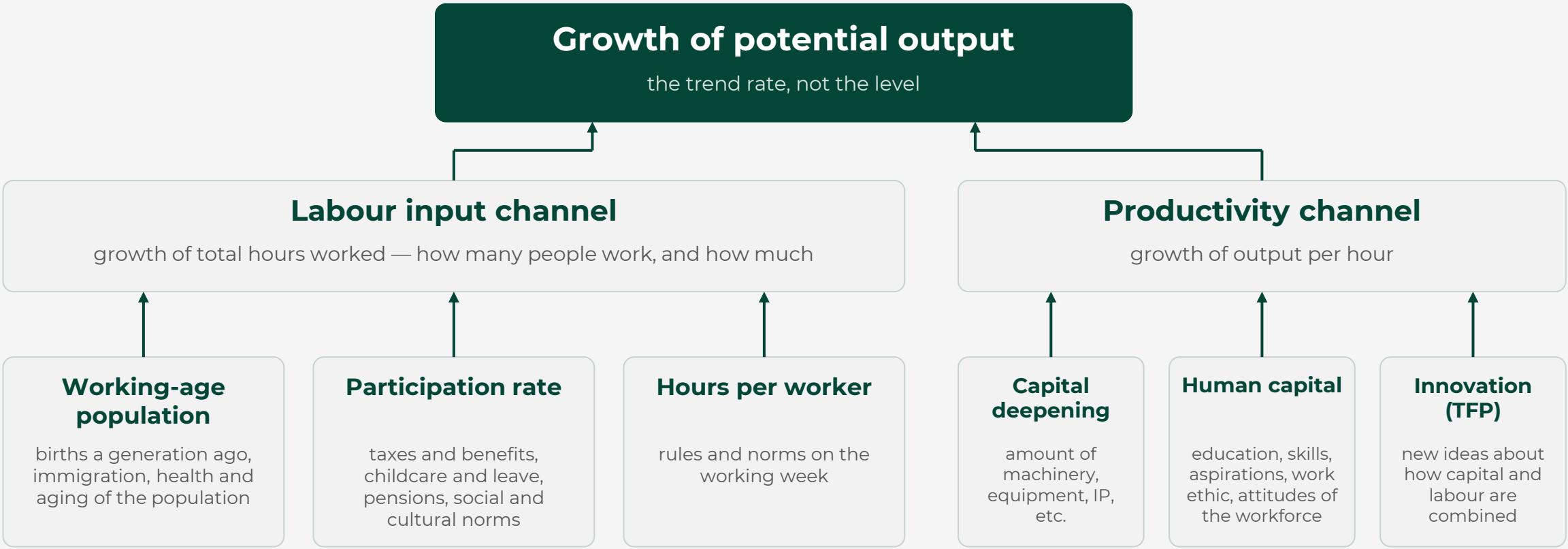
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WARM-UP · HEADLINE 5 OF 9

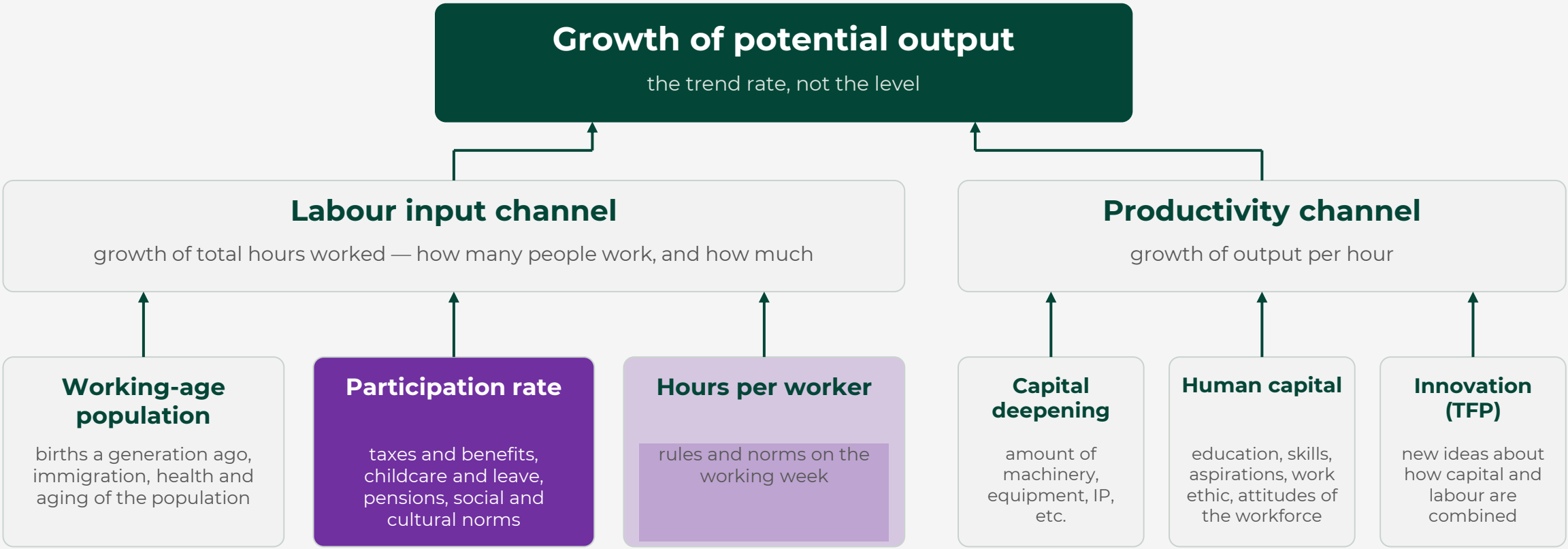
“Ontario increases amount disability recipients keep to $1,000 a month in earnings before benefits shrink.”



Which box does this move — or is it outside the labour input channel altogether?

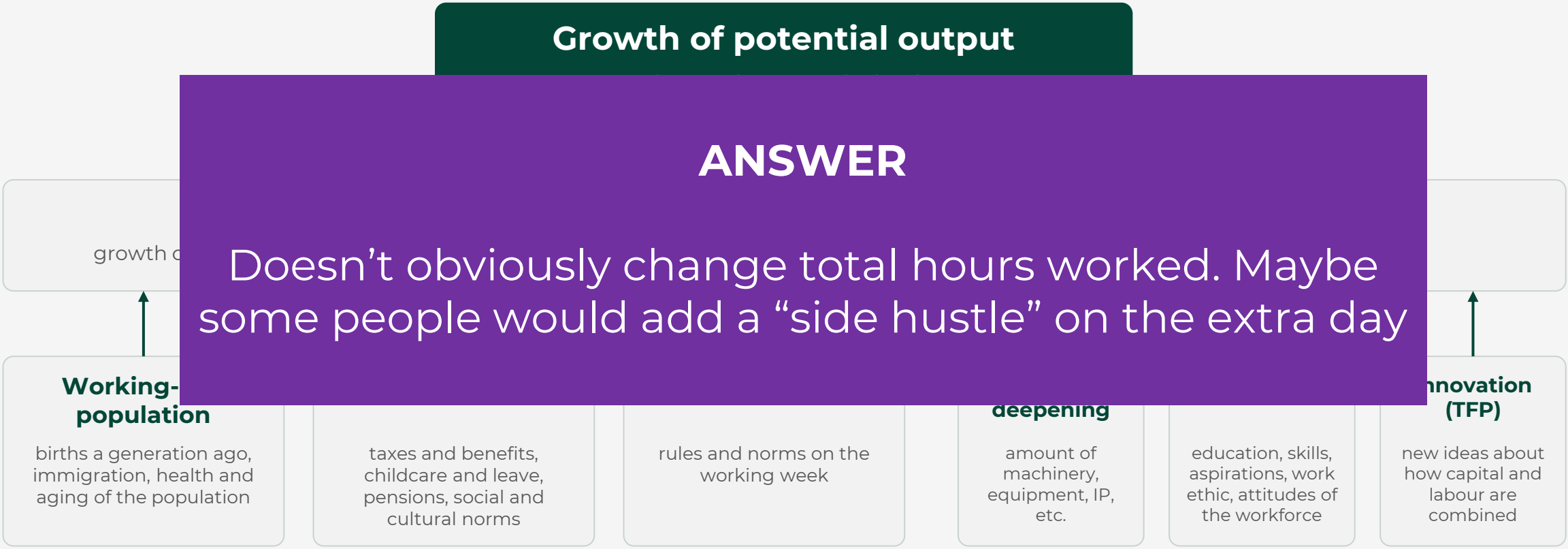
WARM-UP · HEADLINE 5 OF 9

“Ontario increases amount disability recipients keep to $1,000 a month in earnings before benefits shrink.”



Which box does this move — or is it outside the labour input channel altogether?

“Belgium gives workers the right to compress their weekly hours into a four-day week.”



ANSWER

Doesn’t obviously change total hours worked. Maybe some people would add a “side hustle” on the extra day

Which box does this move — or is it outside the labour input channel altogether?

“Shopify to managers: prove AI can't do the job before you hire.”

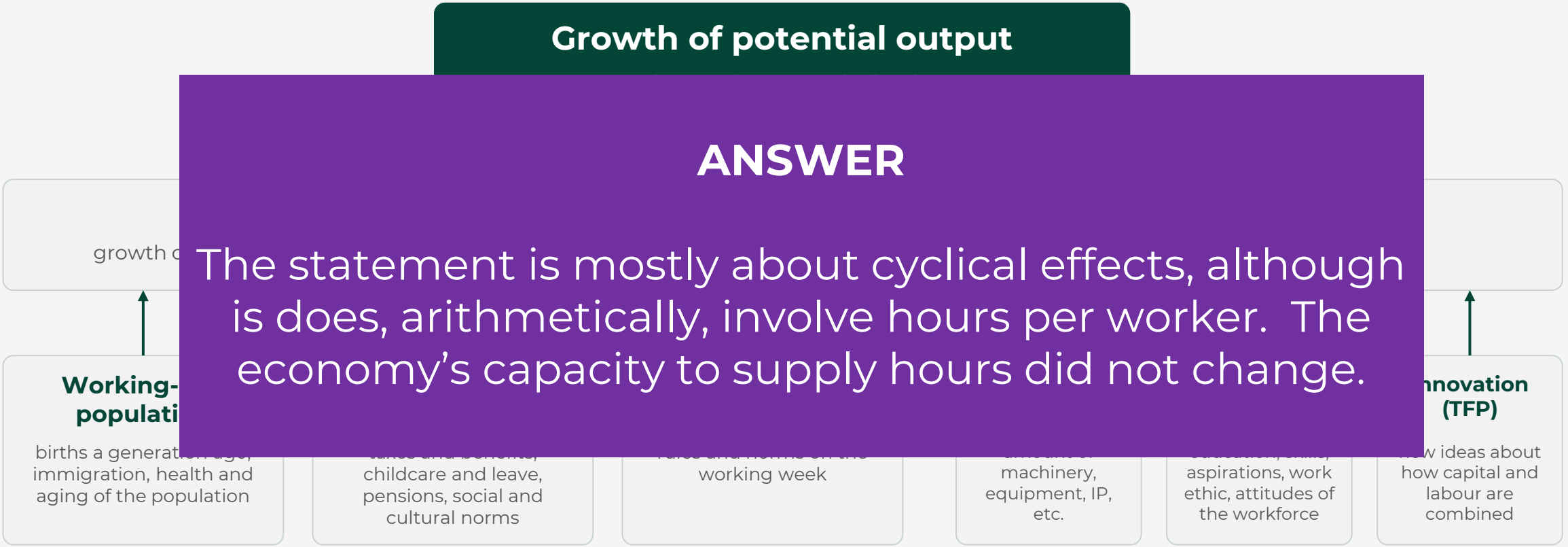


ANSWER

This is productivity channel: capital deepening (investment in software) and innovation (new ways of combining labour and capital)

Which box does this move — or is it outside the labour input channel altogether?

“Overtime in auto plants falls to a 15-year low as tariffs bite.”

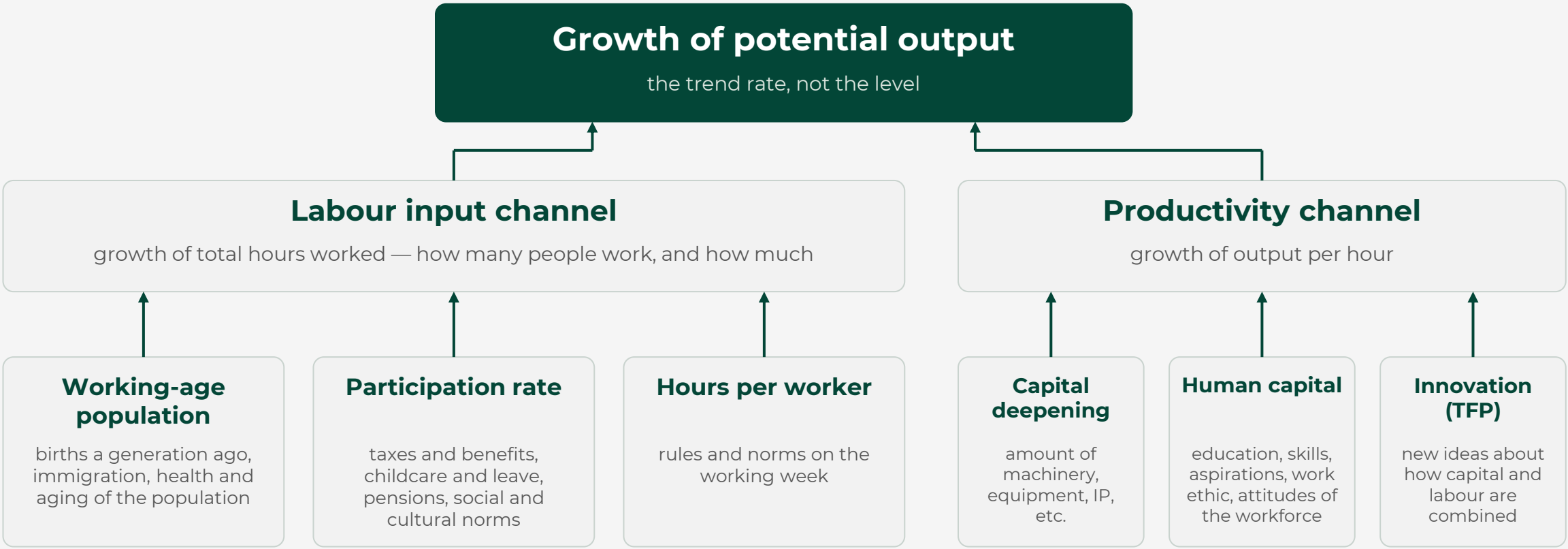


ANSWER

The statement is mostly about cyclical effects, although is does, arithmetically, involve hours per worker. The economy’s capacity to supply hours did not change.

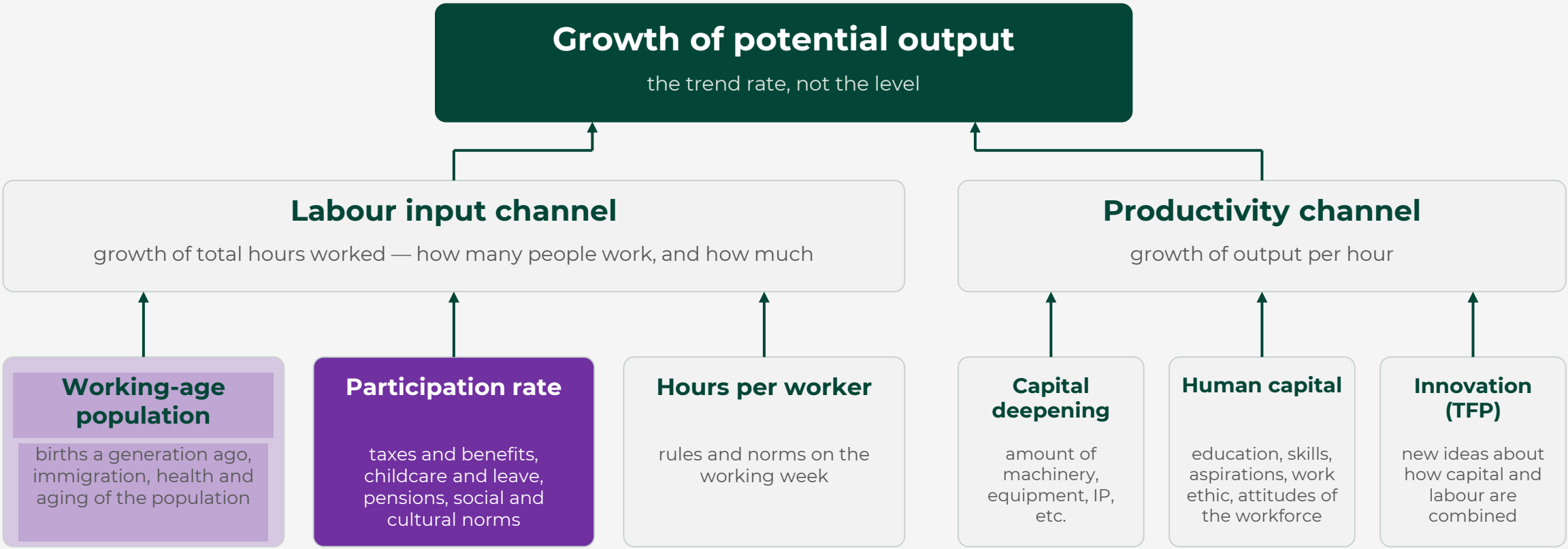
Which box does this move — or is it outside the labour input channel altogether?

“#vanlife: Post-pandemic, millions of Americans are choosing meaning over money, and employers can't find staff.”



Which box does this move — or is it outside the labour input channel altogether?

“#vanlife: Post-pandemic, millions of Americans are choosing meaning over money, and employers can't find staff.”



Which box does this move — or is it outside the labour input channel altogether?

CASE STUDY

Europe: Hours worked

What do you believe? 2 polls on working hours

1 Germany versus Greece

Who works more hours per year? Germans or Greeks?

Germans work more

Greeks work more

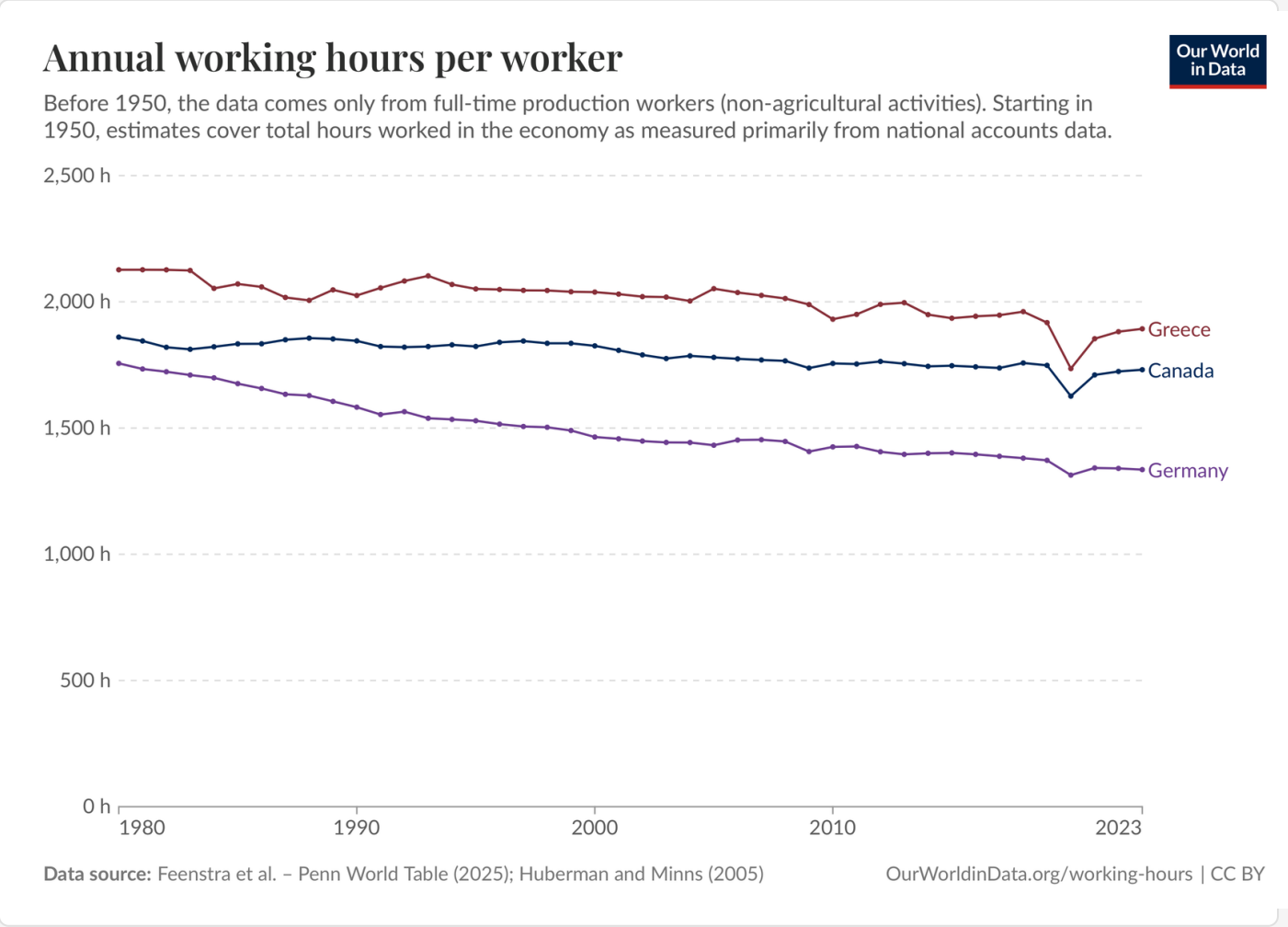
2 Germany versus Canada

Who works more hours per year? Germans or Canadians?

Germans work more

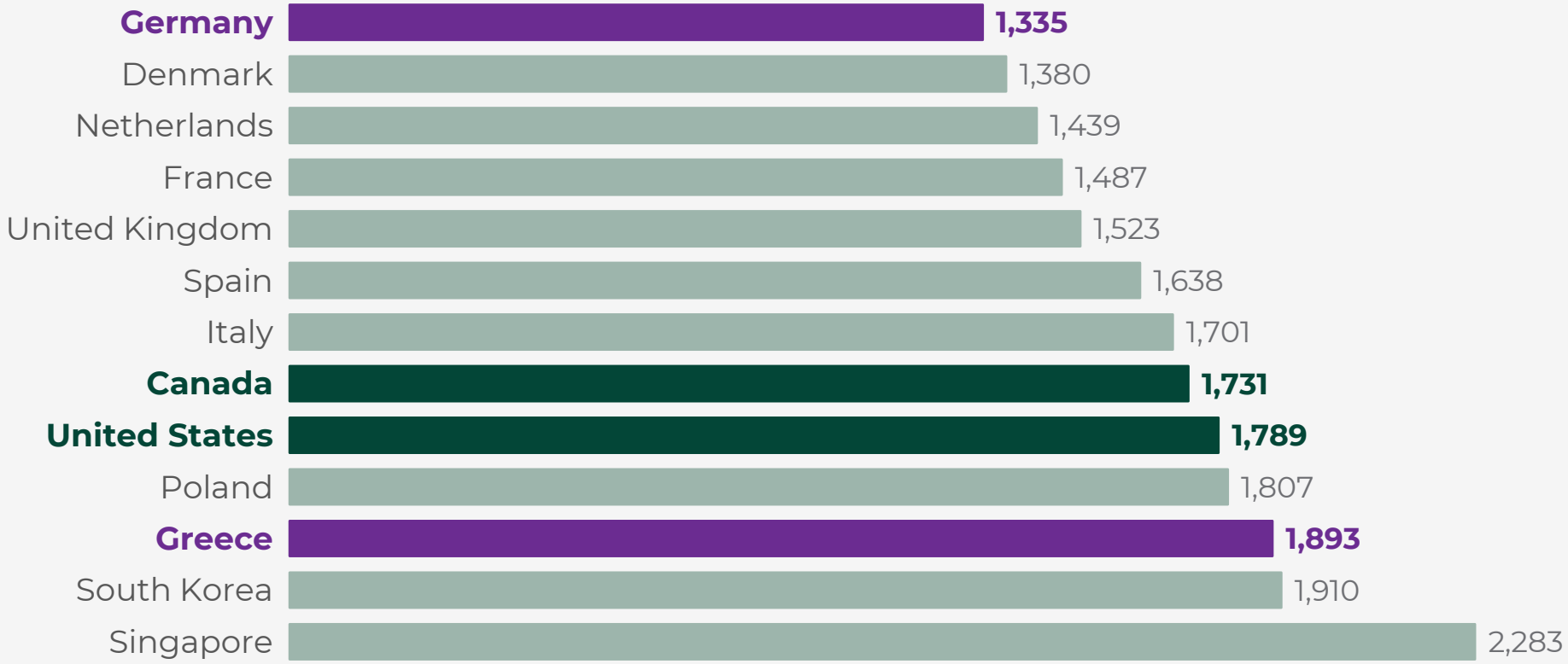
Canadians work more

Κώστα, η πολλή δουλειά και η καθόλου διασκέδαση σε κάνουν βαρετό αγόρι.



Source: Penn World Table via Our World in Data

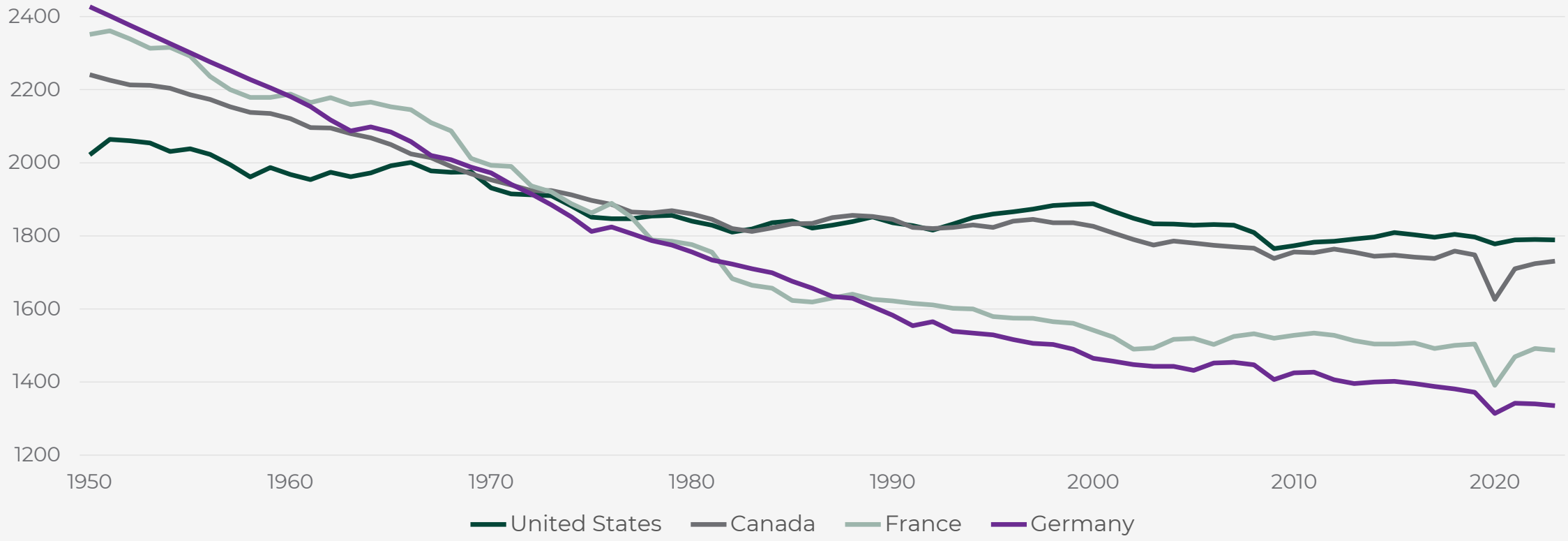
First, “Europe” is not one thing: Greece out-works Canada, and Germany works the least in the rich world



The low-hours club is the rich north-west of Europe. Southern Europe works American hours or more — and produces far less per hour.

Annual hours worked per worker, 2023. Penn World Table 11 via Our World in Data — the source behind Figure 3 in the primer. Mexico omitted: PWT and OECD disagree by 600 hours.

The gap is fifty years old: until the mid-1970s Germans and the French worked more hours than Americans



Any explanation has to name what changed after 1970 — in Europe, and not in the US.

Annual hours worked per worker. Penn World Table 11 via Our World in Data. The 2020 dip is the pandemic.

BUZZ IN PAIRS · 5 MINUTES

Germans work 450 fewer hours a year than Americans. The French, 302 fewer. In 1970 they worked more. Why?

1,335
hours a year per German worker, 2023

1,789
hours a year per American worker, 2023

99%
German output per hour, as a share of American

- Develop three hypotheses that could explain the hours worked gap that opened after 1970.
- For each: what evidence would show it is right — and what would show it is wrong?
- Productivity is off the table. Same output per hour, 450 fewer hours — the primer says productivity permits shorter hours but does not dictate them.

Three hypotheses – and the economists attached to them

TAXES

Edward Prescott (2004)

Effective marginal tax rates on labour rose across Europe after 1970 and stayed at 40 per cent in the United States. Hours fell in step. “Virtually all” of the France–US gap is taxes.

RULES AND UNIONS

Alesina, Glaeser & Sacerdote (2005)

Unions in declining industries bargained “work less, work all”; statutory hours and mandated vacation followed. Then a social multiplier: leisure is worth more when everyone else is off too.

PREFERENCES

Olivier Blanchard (2004)

Productivity levels are similar on both sides of the Atlantic. Europe used the productivity gains of thirty years to buy leisure; the United States used them to buy income.

Digging deeper: the evidence

	FOR	AGAINST
TAXES	Effective marginal tax rates, 1970–74 to 1993–96: France 49 to 59 per cent, Germany 52 to 59, Italy 41 to 64 — the United States 40 both times. Hours fell where taxes rose.	The response required is a labour-supply elasticity far above anything micro studies find. And taxes do not explain why the cut came as six weeks of vacation.
RULES	France cut the legal week from 39 to 35 hours in 2000, taxes unchanged. Hours fell; employment did not rise; turnover rose as firms shed workers who had become dearer.	Laws follow preferences as often as they lead them. Surveys after the 35-hour week: women gained, managers lost. Was it imposed, or was it wanted?
PREFERENCES	Every rich country cut hours as it grew richer (Figure 3). North-western Europe simply went further — and did so together, whatever each country's tax or union history.	"Preferences" is what is left after the other explanations. It predicts nothing that could fail, so it cannot be tested — only assumed.

Hours worked are a fact. How you *judge* differences in hours worked depends on the mechanism at work

Productivity permits shorter hours; it does not dictate them.

Germany and the United States produce the same per hour. Germany works 450 less. Productivity does not decide hours worked. Other things do.

The mechanism decides the verdict – and the response

- A tax distortion is a loss. This can and should be fixed by policy
- An institutional bargain is a collective choice. Poorer in income, but not obviously worse off.
- A preference is a purchase. GDP per person counts the goods and not the time, so the “gap” is a measurement problem, not a distortion.

Where would rather (live)? USA, France, Germany? Explain.

GDP per person = output per hour × hours per worker × workers per capita

	France	Germany
Output per hour (% of US)	93%	99%
× Hours per worker (% of US)	83%	75%
× Share of population in work (% of US)	91%	112%
= GDP per capita	**71%**	**82%**

Each entry is the country's value as a share of the United States, 2023. Penn World Table 11 via Our World in Data; output-side real GDP at 2021 PPP.

Are Europeans poorer than Americans?

1	**Yes — a loss**	Prescott. Taxes make the extra hour not worth working. The shorter hours are a distortion and the 30 per cent is a real income gap, given up for nothing.
2	**They chose it — together**	Alesina, Glaeser & Sacerdote. Unions and laws set the hours; the social multiplier made people like them. The income gap is real, and most would not trade it back.
3	**No — a purchase**	Blanchard. The productivity gains were spent on time instead of goods. GDP per person counts the goods and not the time, so the "gap" is in the ruler – the way we measure "rich" and "poor".

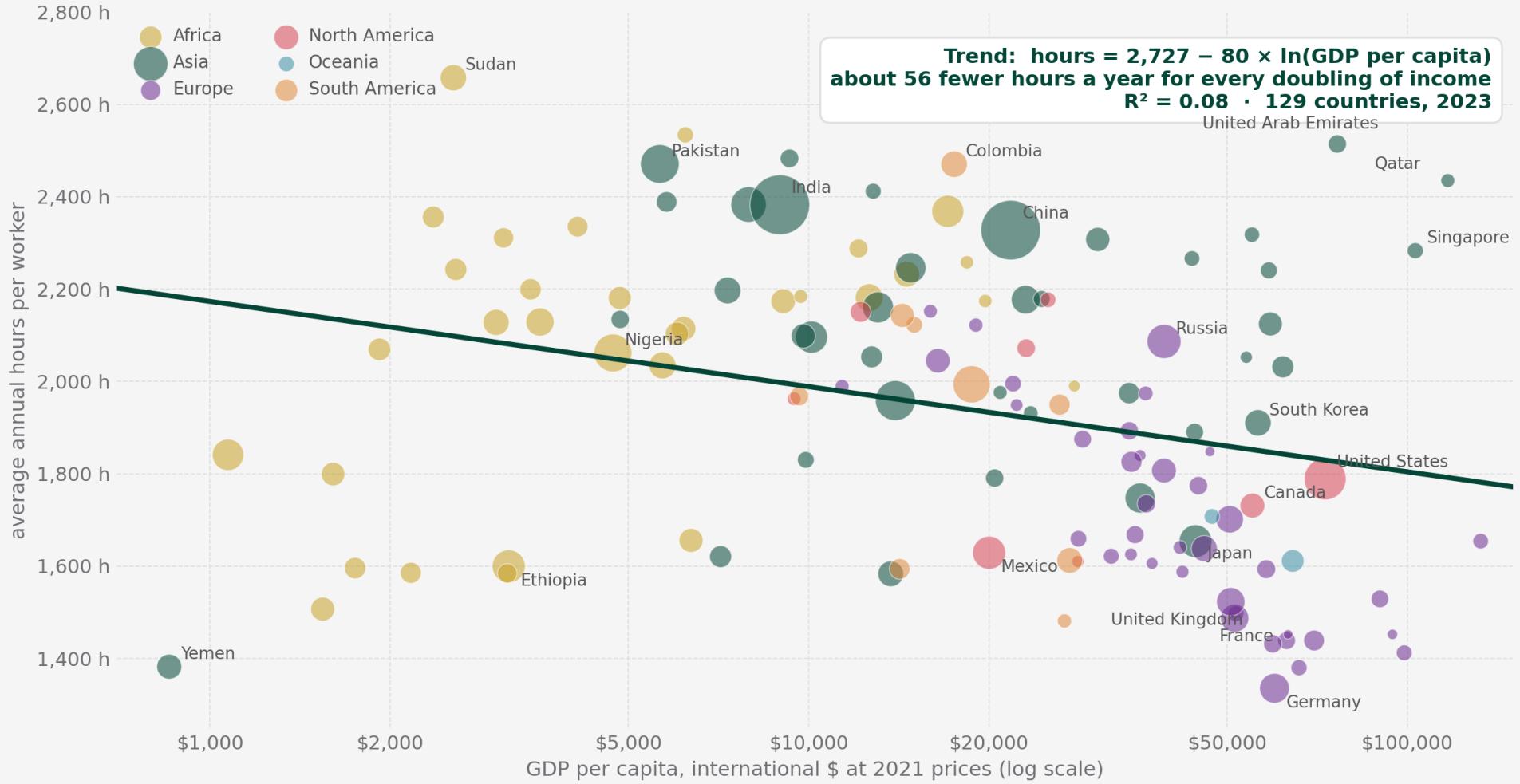
Is it possible that the question is backwards? Maybe Americans are not working "normal" hours? Maybe Americans are working MORE than they would like?

CASE STUDY · THE HOURS MARGIN, INSIDE ONE COUNTRY

The United States

Who works more, the rich or the poor?

First, the global picture: Workers in richer countries tend to work fewer hours than those in poorer countries.



Penn World Table 11 via Our World in Data, 2023. Circles sized by population. OLS on log GDP per capita, unweighted.

POLL

Who works longer hours in the United States — high-income workers, or low-income workers?

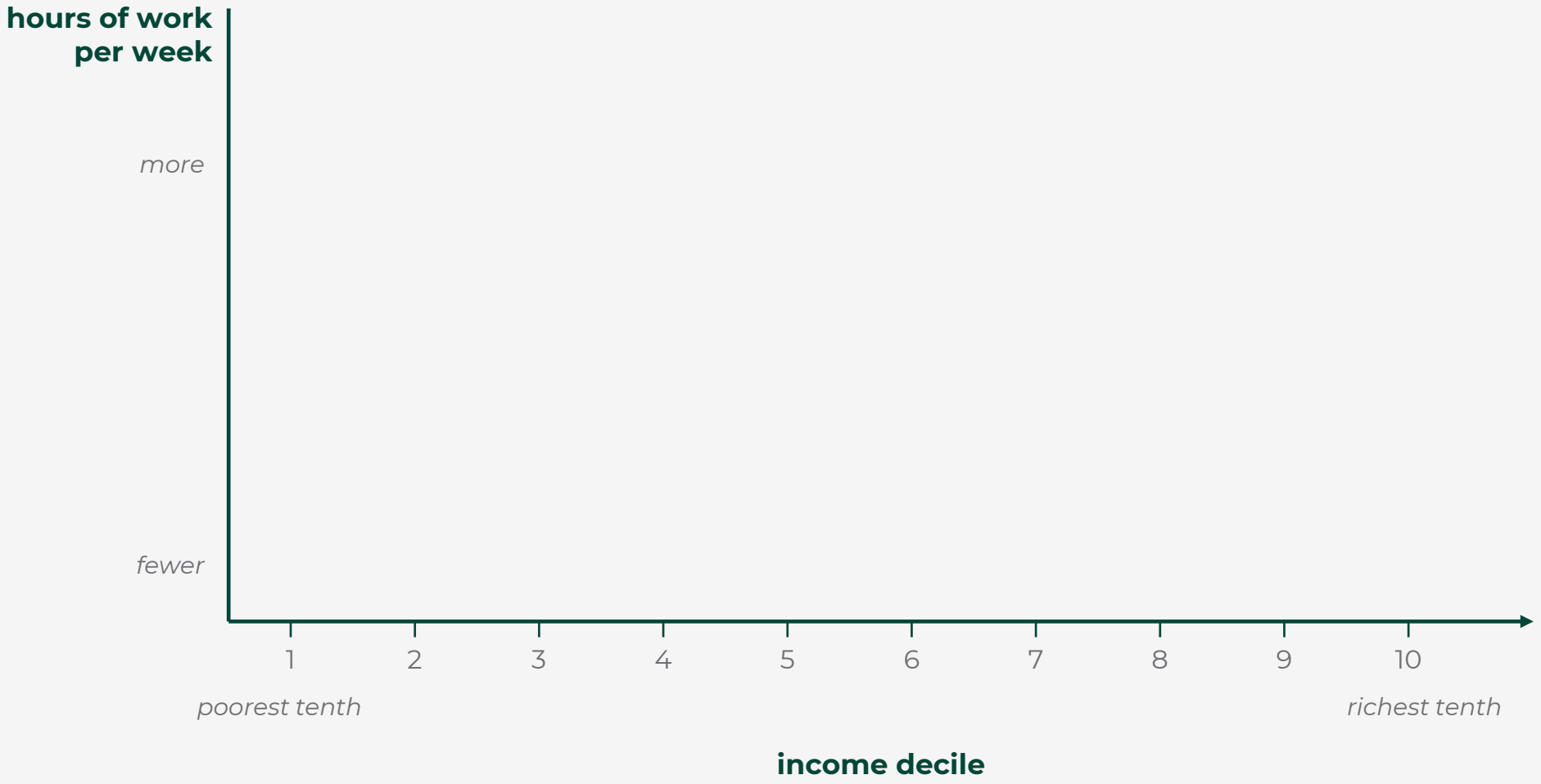
A High-income workers work longer hours

B Low-income workers work longer hours

C About the same

PAIRS · 3 MINUTES

Draw it. Income across, hours of work up. What shape is the line?



ON YOUR SKETCH

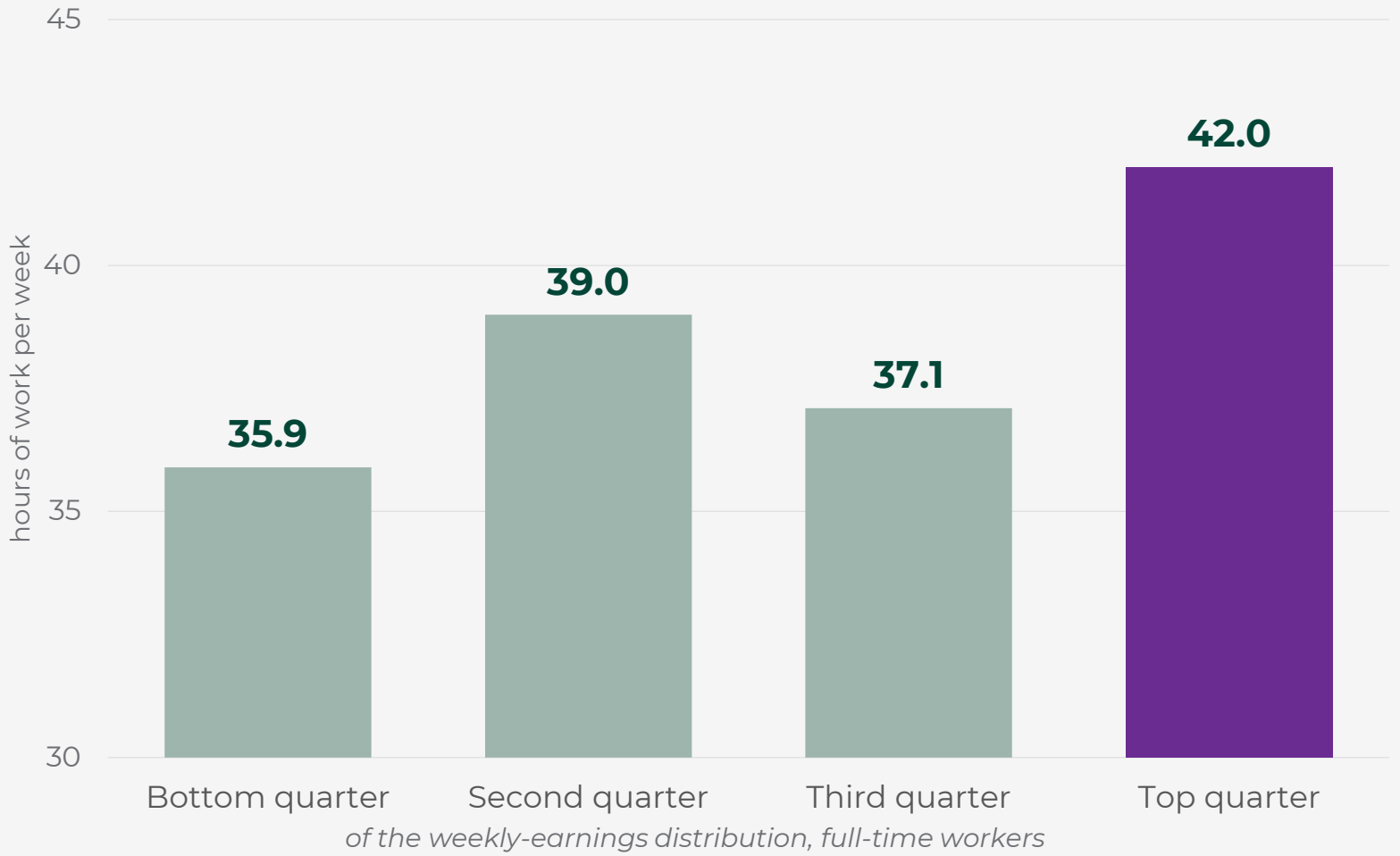
Is it a line, a curve, or a U?

Where is the top? Where is the bottom?

How big is the gap — five hours a week, or twenty?

Does the far right end look like the rest of the line?

The answer: top earners work about six hours a week more than bottom earners



2025

42

hours a week, top quarter of earners

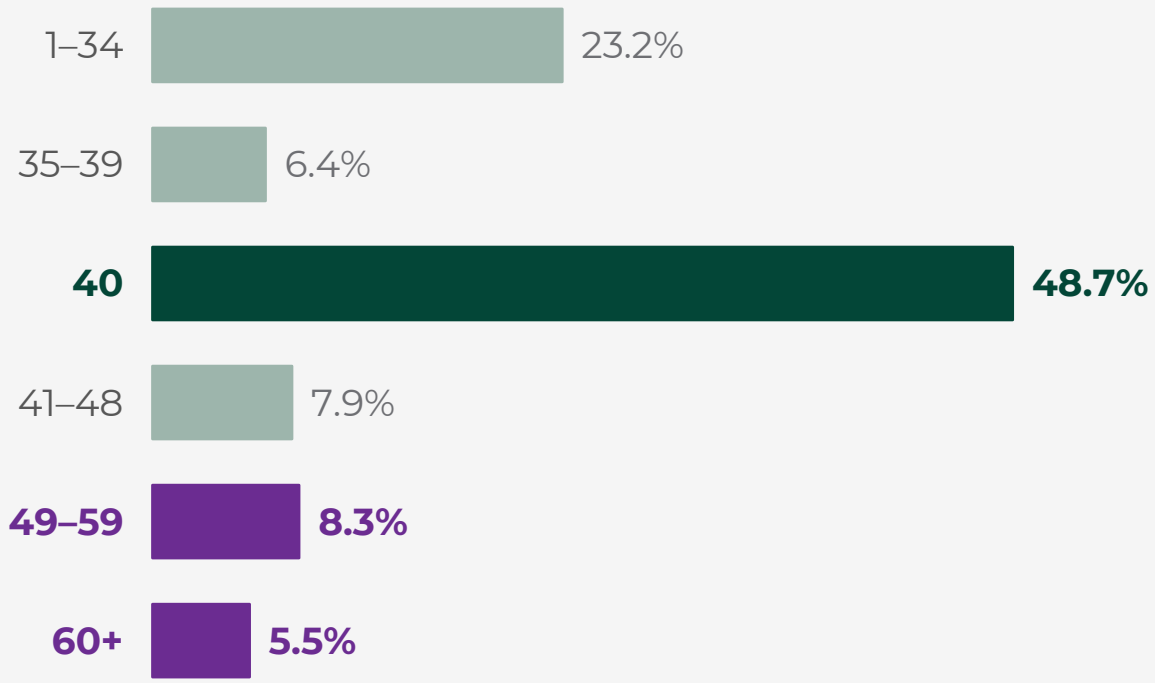
36

hours a week, bottom quarter

Full-time workers only, so this is not part-time versus full-time. The gap has held for twenty years.

American Time Use Survey, Table 5, 2025: full-time wage and salary workers holding one job, by quartile of usual weekly earnings. Weekly hours = share of days worked × hours on days worked × 7.

Averages hide the tail: one in seven works 49 hours or more — and the tail is not a random sample. The tail is your future.



usual hours per week, everyone at work, 2025

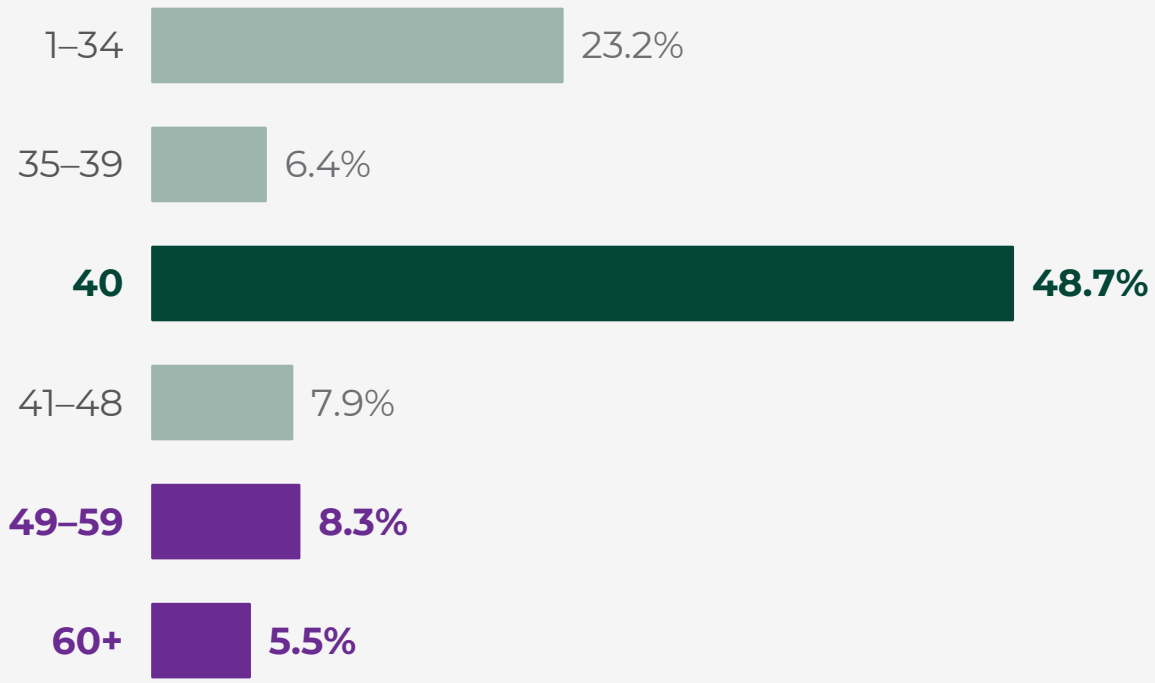
WHO IS IN THE TAIL

30% vs 10% of salaried workers versus hourly-paid workers

18% vs 9% of men at work put in 49+ hours, versus women, 2025

BLS Current Population Survey, Table 19, 2025 annual averages (persons at work, by sex). Kuhn & Lozano (2008), Journal of Labor Economics 26(2), Table 1: employed men 25–64, 2006, by pay basis and by fifth of hourly earnings.

Why do the best paid work the longest — and why is the tail salaried, well paid and skewed male? Hypotheses?



usual hours per week, everyone at work, 2025

WHO IS IN THE TAIL

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BUZZ · 5 MINUTES

BLS Current Population Survey, Table 19, 2025 annual averages (persons at work, by sex). Kuhn & Lozano (2008), Journal of Labor Economics 26(2), Table 1: employed men 25–64, 2006, by pay basis and by fifth of hourly earnings.

Six hypotheses. The first three are about the price of an hour; the rest about rules, status and the work.

1 Substitution beats income

The wage is the price of leisure; raise it enough and people buy less of it. Describes the gradient more than it explains it.

2 Returns to hours are convex

Past 48 hours a week, five extra hours a week buy a point more annual wage growth; below 48, nothing. Goldin's "greedy" jobs: pay per hour rises with hours.

3 Rising inequality raised the prize

Pay at 40 hours stagnated while the premium beyond 40 grew. Long hours rose most where pay inequality rose most. Run harder, for a bigger pot.

4 The salaried exemption

Overtime law makes an hourly worker's 50th hour cost the employer time-and-a-half. A salaried professional's is free. The rise in long hours is almost entirely among the salaried.

5 Status inverted

A busy, overworked life has become an aspirational status symbol in the U.S.: observers infer competence and scarcity. In Italy, the inference runs the other way.

6 The work itself is different

High-paid work is often interesting, autonomous and identity-conferring. Leisure is a closer substitute for a bad job than for a good one.

Hypothesis 2 is the best identified — and it is the one that describes the jobs HBA grads are likely to end up in..

Gicheva (2013), JOLE 31(4); Goldin (2014), AER 104(4); Kuhn & Lozano (2008); Fair Labor Standards Act overtime exemptions; Bellezza, Paharia & Keinan (2017), JCR 44(1).

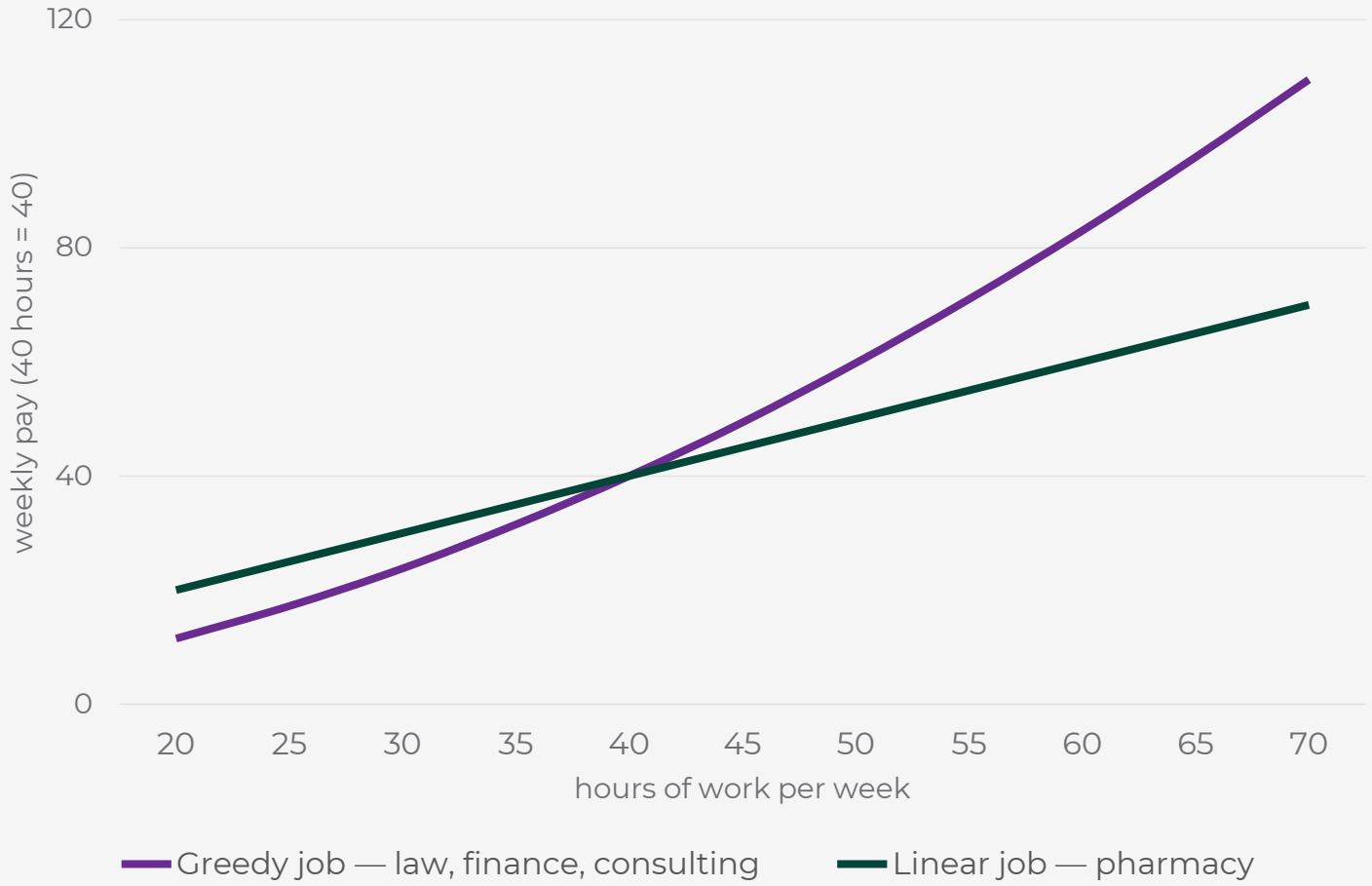


News

Claudia Goldin Wins the Nobel Prize in Economics

Congratulations to Harvard University Press author Claudia Goldin, who has been awarded the Nobel Prize in Economics. Her groundbreaking work as an economic historian and labor economist has advanced our understanding of inequality and women's participation in the workforce.

“Greedy jobs” pay convexly: the 50th hour pays more than the 30th, and part-time pays a penalty



WHAT MAKES A JOB GREEDY

The client owns your time.

The worker is not substitutable: the deal, the case, the patient are yours, and you must be available when they call.

Hours pay more than one-for-one.

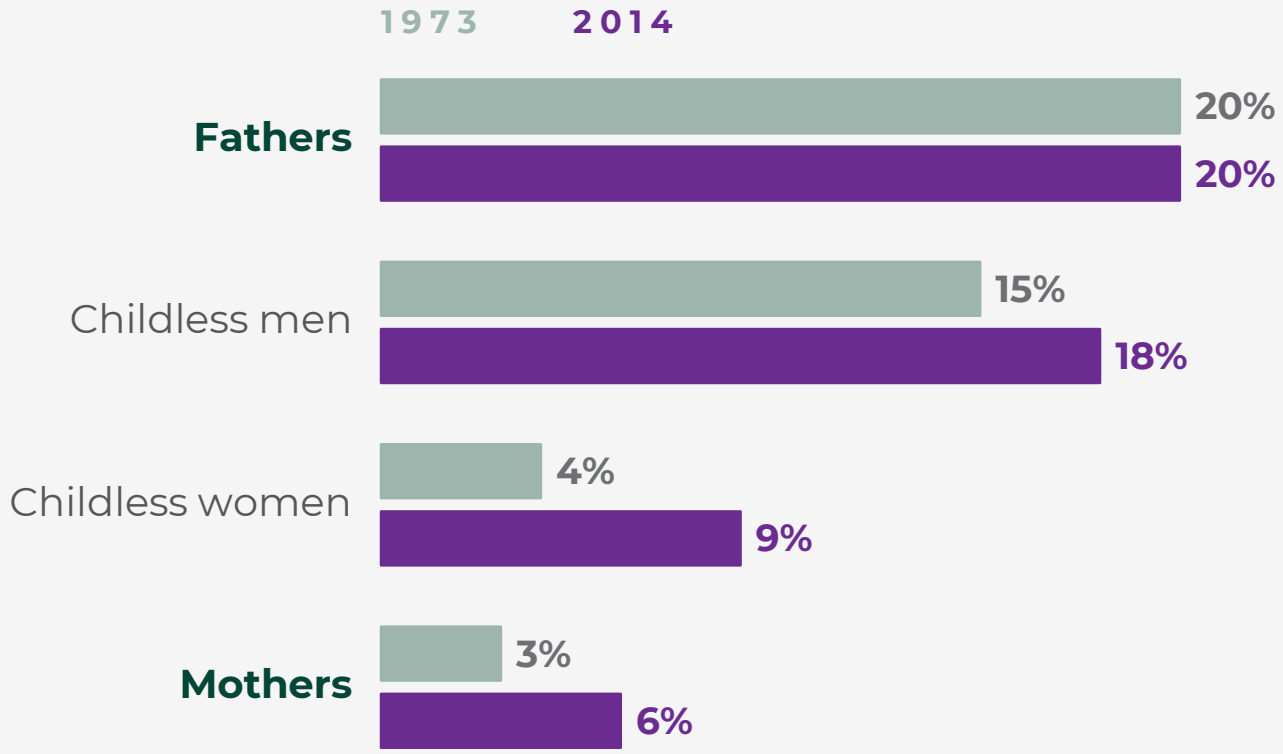
Earnings rise more than one-for-one with hours — and fall more than one-for-one when hours drop. A 30-hour week in a greedy job pays far less than three-quarters.

Substitutable jobs are linear.

Pharmacists hand a prescription to the next pharmacist. Pay per hour is the same at 30 and 50 hours, and the gender pay gap nearly vanishes.

Illustrative curves. Goldin (2014), “A Grand Gender Convergence: Its Last Chapter,” AER 104(4); Goldin, Career and Family (2021); Goldin & Katz (2016) on pharmacy, JOLE 34(3).

Who supplies the 50-hour weeks: fathers most, mothers least — and the ordering has not moved in forty years



share of employed workers usually working 50+ hours a week

THE PRICE OF A 50-HOUR WEEK

−16%

hourly-wage penalty for working 50+ hours, 1979

+6%

hourly-wage premium for working 50+ hours, 2009 — the same for men and women

The price of the long week is the same for everyone. Who can supply it is not.

Weeden, Cha & Bucca (2016), RSF Journal 2(4), CPS. Overwork premium: Cha & Weeden (2014), American Sociological Review 79(3), CPS 1979–2009, 50+ hours vs 35–49, adjusted.

The hours and earnings gap opens with the first child: greedy work pays for hours and punishes interruptions

61 / 59

hours a week, men / women, one year after the MBA

57 / 51

hours a week, nine years after the MBA

−24%

MBA mothers' weekly hours relative to men. Childless women: −3%

$442k / $243k

average earnings, men / women, 10+ years after the MBA

Only half. At 10–16 years out, 52 per cent of MBA mothers work full-time, full-year; 22 per cent of MBA women work part-time.

The interruption tax. Any career break costs about 23 log points of earnings, before the lost experience is counted. Greedy work does not pause.

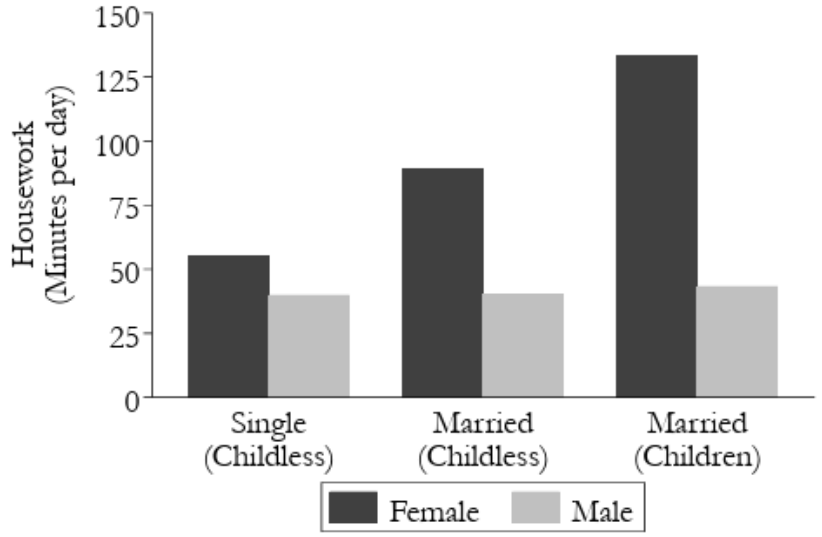
The couple decides, under prices set by the job. Someone must take the flexible job. The long-run child penalty in US earnings is about 31 per cent for mothers and near zero for fathers.

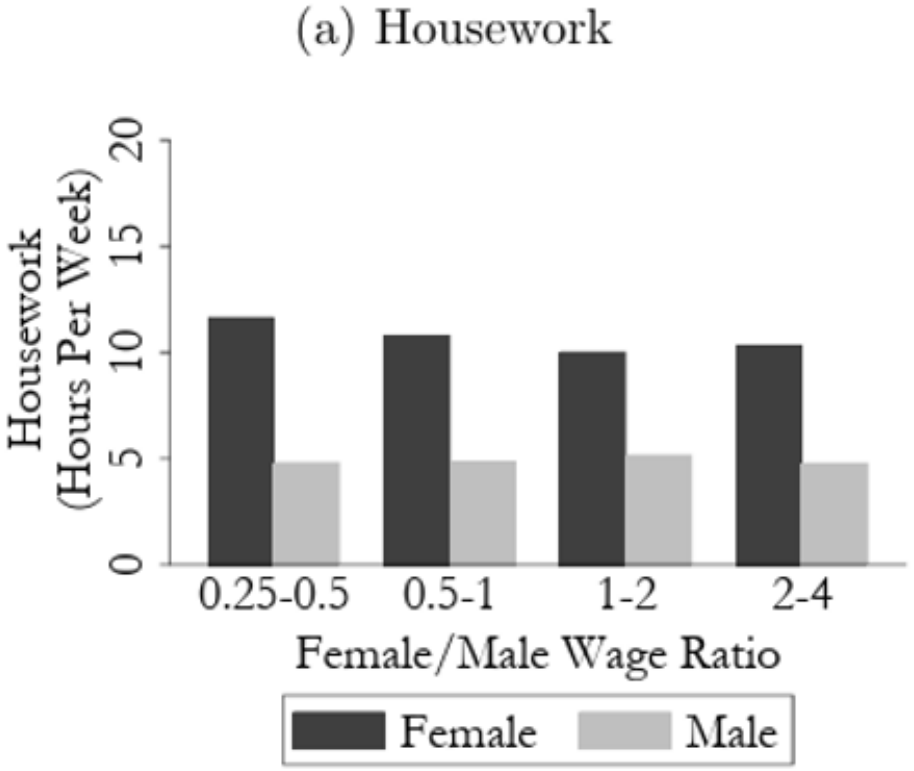
Bertrand, Goldin & Katz (2010), AEJ: Applied 2(3): University of Chicago Booth MBAs, classes of 1990–2006. Child penalty: Kleven, Landais, Posch, Steinhauer & Zweimüller (2019), AEA Papers & Proceedings, years 5–10 after first birth.

The housework holdout

Men's choices set women's limits

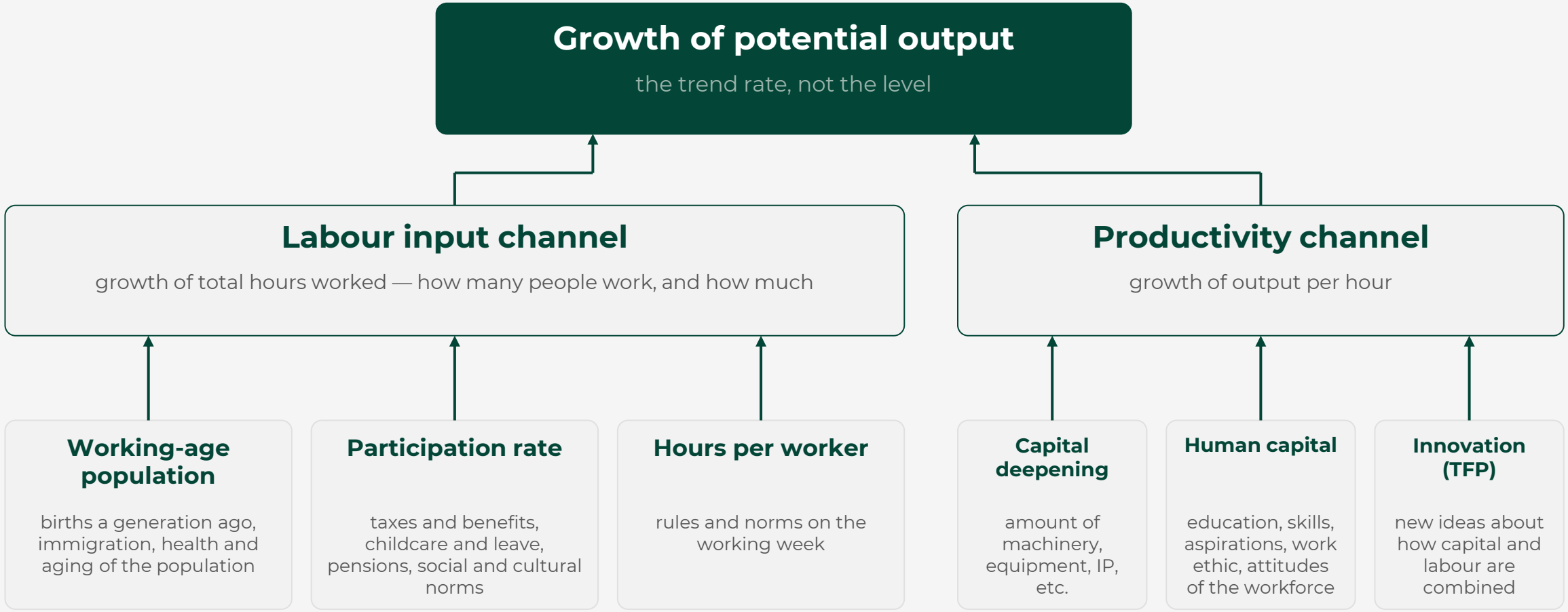
Figure 4: Housework of Men and Women By Marital Status and Fertility





Know thy growth model

We're having super interesting conversations (at least I think so). At the end of it all, bring it back to this



Growth: The Labour Input Channel

Next: the productivity channel.